Retina-Scan: A Deep Learning-Based Retinal Fundus Image Screening Tool for Early Detection of Diabetic Retinopathy

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***Abstract*—Diabetic Retinopathy (DR) is a progressive retinal complication of diabetes in which early manifestations may be difficult to identify through routine manual examination. Retina-Scan is a web-based deep learning screening system designed to support early pre-screening and referral of DR from retinal fundus images. The system follows an end-to-end pipeline consisting of image upload and validation, fundus-image verification using a simple image heuristic, Contrast Limited Adaptive Histogram Equalization (CLAHE), resizing and normalization, EfficientNetB1-based transfer learning, confidence estimation, and Gradient-weighted Class Activation Mapping (Grad-CAM). The classifier is designed to distinguish five severity levels: No DR, Mild, Moderate, Severe, and Proliferative DR. Class-weighted training is incorporated to reduce the influence of class imbalance. The application uses a Flask backend with a browser-based frontend and OpenCV/PyTorch components for image processing and inference. Unlike a purely black-box screening workflow, Retina-Scan also provides a visual heatmap indicating image regions that influenced the predicted class. The project uses APTOS 2019 for model development and evaluation. The project describes the motivation, literature background, architecture, methodology, implementation, evaluation strategy, limitations, and future extensions of the proposed system. The system is intended as a pre-screening and referral-support tool and not as a replacement for professional ophthalmological diagnosis.**

***Index Terms*—Diabetic retinopathy, retinal fundus image, deep learning, EfficientNetB1, transfer learning, CLAHE, Grad-CAM, explainable AI, medical image screening.**

I. Introduction

Diabetic Retinopathy (DR) is a diabetes-related retinal disorder associated with damage to the small blood vessels of the retina. The condition may progress without obvious symptoms during its early stages, making regular screening important. Fundus photography is widely used for retinal screening because it provides a non-invasive representation of retinal structures and lesions. Manual grading, however, requires trained specialists and can become time-consuming when large numbers of images must be reviewed. Recent reviews also identify variability in image quality, class imbalance, generalization, and limited interpretability as important challenges for automated DR systems [2], [10], [11], [14].

Deep learning has changed retinal image analysis by allowing convolutional neural networks (CNNs) to learn discriminative representations directly from images. Studies have demonstrated the usefulness of transfer learning, EfficientNet-family models, and image enhancement techniques for DR classification [1], [4], [5], [10]. EfficientNet is particularly relevant when computational efficiency is important because its scaling strategy jointly considers network depth, width, and input resolution [19]. Recent work has further explored EfficientNet-based models, CLAHE-enhanced fundus images, and explainability mechanisms for practical screening environments [3], [8], [10], [16].

Automated retinal screening research has also expanded toward feature fusion, hybrid deep-learning architectures, and multi-level fundus disease classification. Nur-A-Alam *et al.* investigated fused features from retinal images for automated DR detection [12], while Mutawa *et al.* explored a hybrid CNN-RBF framework for DR screening [13]. More recent studies have investigated multi-level disease identification and broad-coverage classification using fundus images [15], [18]. These studies demonstrate the continued development of automated retinal-image analysis beyond conventional CNN classification.

Retina-Scan is developed around this direction. The intended workflow is explicitly limited to pre-screening and referral support rather than definitive diagnosis. The system also specifies APTOS 2019 as the dataset for model development and evaluation. The design emphasizes a complete application workflow rather than treating the classifier as an isolated component. This includes input validation, preprocessing, classification, confidence reporting, visual explanation, and browser-based result presentation. Such integration is intended to make the model easier to demonstrate and evaluate as an end-to-end screening application.

The main contributions of this work are: (1) an end-to-end web-based DR pre-screening architecture; (2) a five-class EfficientNetB1 transfer-learning classifier with class-weighted training; (3) a contrast enhancement stage based on CLAHE; (4) a simple fundus-image verification heuristic before model inference; (5) an explainability stage using Grad-CAM; and (6) a modular implementation using Flask, PyTorch, OpenCV, HTML, CSS, and JavaScript. The project focuses on the defined design and does not introduce experimental performance values that have not been reported.

II. Related Work

A. Deep Learning for DR Classification

Early automated DR systems commonly relied on image processing, handcrafted features, segmentation, and conventional classifiers. Bilal *et al.* proposed a hybrid framework combining preprocessing, feature extraction, and multiple classifiers with majority voting, demonstrating the value of combining complementary models for disease grading [6]. Such pipelines can reduce dependence on purely manual inspection, but handcrafted feature design and multiple processing stages may increase system complexity.

Khan *et al.* proposed VGG-NiN, which combines VGG16, spatial pyramid pooling, and Network-in-Network components to improve nonlinear representation and scale handling for DR classification [9]. Their work illustrates the transition from conventional feature engineering toward deeper learned representations. AbdelMaksoud *et al.* investigated transfer learning with a customized EfficientNet-based system and reported the usefulness of transfer learning when training data are limited [7].

Recent research has continued to examine automated DR detection using different deep-learning and feature-extraction strategies. Nur-A-Alam *et al.* proposed a Faster RCNN-based method using fused features extracted from retinal images for automated DR detection and lesion identification [12]. Mutawa *et al.* developed a randomization-driven hybrid deep-learning framework combining multi-scale image features with a CNN-RBF classifier for DR screening [13]. These approaches demonstrate the importance of preprocessing, feature extraction, and robust classification in automated retinal screening.

Abu-Shawish *et al.* provided a comprehensive review of deep-learning approaches for automatic DR detection and grading, covering CNN architectures, transfer learning, hybrid models, and Grad-CAM-based interpretation [10]. Islam *et al.* reviewed AI-based retinal health screening using digital fundus images and highlighted the extensive use of CNN and machine-learning approaches for DR and other retinal diseases [11]. These reviews reinforce the importance of automated screening while identifying interpretability, generalization, and image-quality issues as continuing challenges.

B. EfficientNet and Transfer Learning

EfficientNet was introduced as a family of convolutional networks that use compound scaling rather than independently increasing depth, width, or resolution [19]. Its efficiency makes the family attractive for applications where model size and inference cost matter. Momeni Pour *et al.* combined CLAHE preprocessing with EfficientNetB5 and evaluated cross-dataset DR monitoring, showing that contrast enhancement and EfficientNet-based classification can be useful together [4]. Wang *et al.* also used EfficientNet as a feature extractor for fundus-image classification and combined transfer learning with ensemble strategies to address limited training data [1].

A 2022 review of transfer learning for medical image classification found that transfer learning can address data scarcity while reducing training time and hardware requirements [5]. A survey focused specifically on DR classification further highlights supervised deep learning, transfer learning, and newer architectures as major research directions, while noting persistent issues involving dataset variation, interpretability, and generalization [2]. These observations support the choice of EfficientNetB1 transfer learning for Retina-Scan.

Recent work further supports the use of contrast enhancement with EfficientNet-based retinal-image analysis. Ahmad and Alshammari investigated CLAHE-enhanced fundus images with EfficientNet and other pretrained models, including EfficientNetB1, for multiclass ocular disease classification [16]. Their work provides additional support for combining CLAHE-based preprocessing with EfficientNet-family models in fundus-image applications.

C. Explainability in Medical Imaging

Deep neural networks can achieve strong predictive performance while remaining difficult for users to interpret. Grad-CAM addresses this issue by using class-specific gradients flowing into convolutional feature maps to produce a localization heatmap [20]. Recent DR research has explicitly investigated explainable AI to improve transparency and clinician understanding of automated predictions [3], [10]. Ciaramella *et al.* further investigated Grad-CAM, Grad-CAM++, and Score-CAM for explainable DR classification and localization [17]. Retina-Scan incorporates Grad-CAM as a required output rather than treating explainability as an optional visualization.

For a screening-oriented application, explainability is particularly useful because the predicted class alone does not show how the model reached its decision. A heatmap can provide an additional visual reference for the user by indicating the regions receiving stronger model attention. However, the visualization remains a model-derived explanation and should not be interpreted as a clinically verified lesion map. Thus, Retina-Scan treats explainability as supporting information that accompanies the prediction rather than replacing clinical examination.

III. Problem Definition and System Objectives

Manual DR screening depends on expert examination of retinal fundus photographs. The system identifies specialist dependence, time-consuming analysis, limited accessibility in areas with fewer ophthalmologists, difficulty in large-scale screening, and limited automated support for explaining classification decisions as key limitations. The problem addressed by Retina-Scan is therefore the design of an accessible automated pre-screening system that can classify retinal images by DR severity and provide an understandable visual explanation.

The system objectives are to automate DR pre-screening, classify images into No DR, Mild, Moderate, Severe, and Proliferative DR, use EfficientNetB1 for classification, enhance images using CLAHE, reduce class-imbalance effects through class-weighted training, provide a confidence score, generate Grad-CAM explanations, verify that an uploaded image is likely to be a retinal fundus image before inference, provide a

TABLE I
SUMMARY OF SELECTED LITERATURE AND THEIR RELEVANCE TO RETINA-SCAN

Work	Main approach	Relevance to proposed system
Wang *et al.* [1]	EfficientNet + transfer/ensemble learning	Supports learned feature extraction and limited-data fundus classification
Atwany *et al.* [2]	DR deep-learning survey	Identifies model families, datasets, and open challenges
Shahzad *et al.* [3]	Explainable DR diagnosis	Motivates interpretable model outputs
Momeni Pour *et al.* [4]	CLAHE + EfficientNetB5	Supports contrast enhancement and EfficientNet-based DR analysis
Kim *et al.* [5]	Transfer-learning review	Supports pretrained models for medical-image data scarcity
Bilal *et al.* [6]	Mixed-model DR classification	Demonstrates automated disease grading approaches
AbdelMaksoud *et al.* [7]	Transfer learning + EfficientNet	Supports transfer learning for DR grading
Dharrao *et al.* [8]	EfficientNetB0 + attention + Grad-CAM	Supports lightweight, interpretable screening direction
Khan *et al.* [9]	VGG16 + SPP + NiN	Demonstrates deep feature learning for DR classification
Abu-Shawish *et al.* [10]	DR deep-learning survey + Grad-CAM	Supports CNNs, transfer learning, grading, and explainability
Islam *et al.* [11]	AI retinal screening review	Supports fundus-image screening and future directions
Nur-A-Alam *et al.* [12]	Faster RCNN + fused features	Supports automated DR detection and retinal feature extraction
Mutawa *et al.* [13]	Hybrid CNN-RBF	Demonstrates alternative deep-learning screening architecture
Soomro *et al.* [14]	Retinal image analysis survey	Supports DR grading and image-quality considerations
Kulkarni *et al.* [15]	OGFA + CNN	Supports multi-level fundus disease identification
Ahmad *et al.* [16]	CLAHE + EfficientNet/GCViT	Strong relevance to CLAHE and EfficientNet-based fundus classification
Ciaramella *et al.* [17]	Grad-CAM variants + DR localization	Supports explainable DR classification and localization
Rieck *et al.* [18]	Transformer-CNN + fundus images	Supports robust fundus screening and clinical decision support

simple web interface, and support early screening and referral prioritization.

The objectives also emphasize modularity and practical deployment. The image-validation stage is separated from the disease-classification stage so that unsuitable inputs can be rejected before computationally expensive inference. Similarly, the explanation stage is separated from the classification output so that prediction and visualization can be independently tested. This modular organization provides a clear mapping between the project requirements and the software implementation.

IV. PROPOSED SYSTEM

A. System Overview

Retina-Scan is a web-based deep learning application. A user or technician uploads a retinal image through the browser. Before model inference, the Flask backend performs file validation and applies a simple image heuristic to determine whether the uploaded image has characteristics consistent with a retinal fundus photograph. The heuristic considers general visual properties such as image darkness, corner regions, color distribution, and central brightness. If the image does not satisfy the basic fundus-image conditions, it is rejected instead of being passed to the DR classifier.

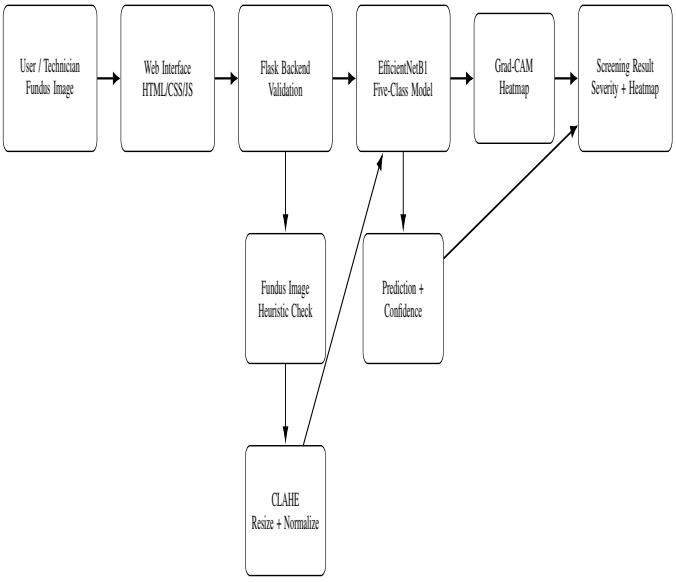


Fig. 1. High-level architecture of Retina-Scan.

For an accepted image, the Flask backend forwards the image to the preprocessing pipeline. CLAHE improves contrast, after which the image is resized and normalized to meet the classifier input requirements. The trained EfficientNetB1 model produces class probabilities for the five DR categories. The highest-probability class is selected as the predicted severity and its probability is reported as the confidence score. Grad-CAM then generates a heatmap showing image regions that contributed to the selected prediction. The severity, confidence, original image, and heatmap are returned to the web interface.

The architecture separates input validation from clinical classification. This separation is useful because a classifier trained on retinal images may still generate a class for an unrelated image if such an image is presented to it. The heuristic therefore serves as a preliminary input-quality safeguard rather than a second DR classifier. The overall design consequently contains both a basic input gate and a learned disease-classification stage.

B. Processing Workflow

The operational workflow consists of eight major stages:

1) **Image upload and validation:** The interface accepts a retinal image and rejects unsupported or invalid files.
2) **Fundus-image verification:** A simple image heuristic checks basic visual characteristics of the uploaded image to determine whether it is likely to be a fundus photograph.
3) **Image enhancement:** CLAHE is applied to improve local contrast and retinal-feature visibility.
4) **Resizing and normalization:** The enhanced image is transformed into the input representation required by EfficientNetB1.
5) **Classification:** EfficientNetB1 performs five-class inference using transfer-learned visual features.

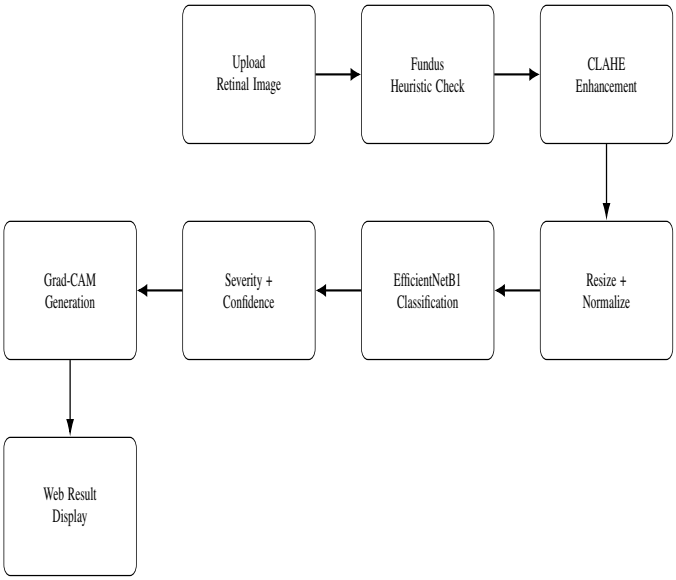


Fig. 2. Retina-Scan processing workflow with fundus-image verification.

6) **Confidence estimation:** The highest predicted class probability is reported as the confidence score.
7) **Explainability:** Grad-CAM uses feature maps and class-specific gradients to generate a heatmap.
8) **Result presentation:** The predicted severity, confidence score, original image, and heatmap are displayed for screening and referral support.

The workflow is designed so that each stage produces an output required by the next stage. File validation prevents unsupported inputs from entering the pipeline, while the heuristic provides a basic fundus-image check. CLAHE and the subsequent transformations prepare the accepted image for the learned model. Finally, the prediction and explanation are combined at the presentation layer so that the user receives both the classification result and its visual interpretation.

V. Methodology

A. Dataset Basis

The project identifies the APTOS 2019 dataset as the project dataset. APTOS 2019 is a diabetic-retinopathy grading dataset commonly used for five-level retinal-image classification. The literature emphasizes that dataset size, class distribution, and domain variation strongly affect medical-image model performance [2], [5], [11], [14]. Therefore, the project treats dataset quality and representativeness as an explicit system constraint.

The use of more than one dataset is also relevant to the generalization problem. Retinal images can differ because of camera characteristics, illumination, acquisition protocols, image resolution, and population differences. A model that performs well on one dataset may not necessarily demonstrate the same behavior on another dataset. Consequently, the project considers cross-dataset evaluation an important future validation step rather than assuming that performance on one dataset automatically represents broader clinical performance.

B. Fundus Image Verification

An additional validation stage is introduced before DR classification. The purpose of this stage is to prevent obviously unrelated images from being processed by the trained DR classifier. Instead of using another trained neural network, Retina-Scan employs a simple image heuristic based on general visual characteristics of fundus photographs.

The heuristic examines properties such as dark corner regions, color distribution, red-channel dominance, and the brightness of the central image region. Fundus photographs commonly contain a circular retinal field surrounded by darker regions, together with characteristic color and illumination patterns. If the uploaded image does not satisfy the required basic conditions, the system reports that the image does not appear to be a valid fundus image and stops the inference pipeline.

This check is intended only as an input-validation mechanism. It does not guarantee that an image is clinically valid or that it contains a particular retinal disease. Images that pass the heuristic are subsequently processed by the DR classifier. The heuristic therefore acts as a lightweight preliminary filter rather than a replacement for clinical image-quality assessment.

C. Preprocessing with CLAHE

Retinal photographs can contain illumination variation, low contrast, and other acquisition-related differences. Retina-Scan uses CLAHE as its image-enhancement operation. Unlike global histogram equalization, CLAHE operates on local regions while limiting contrast amplification. Momeni Pour *et al.* demonstrated the use of CLAHE before EfficientNet-based DR classification [4]. More recent work has also examined CLAHE-enhanced fundus images with EfficientNet-family models for multiclass ocular disease classification [16]. In Retina-Scan, the enhanced image is subsequently resized and normalized before model inference.

Conceptually, for a local image tile with histogram, CLAHE clips histogram bins at a selected limit and redistributes the clipped mass before applying local equalization. The resulting operation aims to enhance local retinal structures while controlling excessive amplification of noise. The exact implementation parameters are part of the trained preprocessing configuration and are not specified numerically in the current system.

The preprocessing operation must remain consistent between model development and deployment. Applying different resizing, normalization, or contrast-enhancement procedures during inference can introduce a distribution mismatch between training and testing images. Therefore, the deployed Flask application follows the preprocessing representation expected by the trained EfficientNetB1 model.

D. EfficientNetB1 Transfer Learning

EfficientNet models use compound scaling to balance network depth, width, and image resolution [19]. Retina-Scan selects EfficientNetB1 as the backbone because the project requires a practical classifier that can support web-based

inference while providing a larger representation capacity than EfficientNetB0. The project specifies transfer learning rather than training a complete network from scratch. A pretrained representation is adapted to the five-class DR task, allowing the model to reuse visual features while learning the target classification problem.

The PyTorch implementation uses an EfficientNetB1-based architecture with an additional attention component in the classification pipeline. The model is trained using GPU-supported PyTorch when suitable hardware is available. The final classifier outputs a probability vector where the five indices correspond to No DR, Mild, Moderate, Severe, and Proliferative DR. The predicted class is the class with the highest probability, and the displayed confidence is the probability associated with that predicted class. The confidence score expresses model probability and is not interpreted as a clinical certainty measure.

Recent literature also demonstrates continued investigation of EfficientNet-based models for fundus-image classification and automated screening. Wang *et al.* used EfficientNet for fundus-image classification [1], while Ahmad and Alshammari evaluated EfficientNetB1 among several pretrained architectures on CLAHE-enhanced fundus images [16]. These studies provide additional background for the EfficientNetB1 choice in Retina-Scan.

The transfer-learning strategy also provides a practical development advantage because the model begins with representations learned from large-scale image data. The final layers can then be adapted to the retinal classification task. This approach reduces the need to learn all low-level visual features from the available retinal datasets and is suitable for a project where computational resources and training time are constrained.

E. Class-Weighted Training

The system explicitly identifies class imbalance as a training concern and requires class-weighted training. Class weights are assigned to the different severity categories and incorporated into the classification loss during model training. The purpose is to prevent classes with fewer training examples from contributing disproportionately little to the optimization objective. The exact class weights are implementation parameters and are not prescribed by the project.

Class weighting is particularly relevant to multi-class DR grading because classification errors are not necessarily distributed uniformly across the five categories. Without appropriate treatment, a model may become more influenced by classes containing larger numbers of training examples. Weighting the loss provides a mechanism for giving greater importance to underrepresented categories during optimization while retaining the same five-class prediction structure.

F. Grad-CAM Explainability

Grad-CAM produces a class-specific localization map from the gradients of a target class with respect to convolutional feature maps [20]. The importance of each feature map is determined from the gradients associated with the target class. The resulting coarse localization map is then generated from the weighted feature maps.

The map is resized and overlaid on the retinal image to indicate regions that contributed to the prediction. In Retina-Scan, this heatmap is displayed with the severity and confidence score so that the result is not presented as an unexplained label. Recent work in DR has also evaluated Grad-CAM and related class-activation mapping methods for explainable classification and localization [3], [17]. This supports the inclusion of visual explanations as part of the screening workflow.

The Grad-CAM output is generated after the predicted class has been identified. This ensures that the visualization corresponds to the class presented to the user. The resulting heatmap is intended to support visual inspection of the model's attention and to improve transparency. It should nevertheless be considered an explanatory visualization rather than an independent diagnostic measurement.

VI. Implementation Architecture

A. Web Application Layer

The frontend is implemented using HTML, CSS, and JavaScript and provides image upload, fundus-image validation feedback, and result presentation. The interface is designed for a modern web browser and focuses on a simple screening interaction. The output contains the predicted severity, confidence score, original retinal image, and Grad-CAM visualization.

The frontend communicates with the Flask backend through the defined application routes. User actions such as selecting an image and submitting it initiate the corresponding backend processing. Validation or inference errors are returned to the interface in a controlled manner so that the user can understand whether the problem occurred during file validation, fundus verification, or model processing. This design keeps the user interaction simple while retaining the modularity of the backend.

B. Backend and Inference Layer

Flask provides the application backend. It receives the uploaded image, performs file validation, applies the fundus-image heuristic, invokes the preprocessing pipeline, loads the trained EfficientNetB1 model, performs inference, generates the confidence value and Grad-CAM output, and returns the result to the frontend. PyTorch provides the deep-learning runtime, while OpenCV supports image processing and the fundus-image heuristic. NumPy and related Python components are used for image and model processing. GPU-enabled environments can be used for model development and training.

The inference layer is designed to load the trained model once during application startup where appropriate, rather than repeatedly constructing the network for every request. The image-processing stages are kept separate from the model definition so that changes to the web interface do not require

TABLE II
CORE RETINA-SCAN MODULES

Module	Responsibility
Image Upload	Accept and validate retinal fundus images; reject invalid formats.
Fundus Verification	Use a simple visual heuristic to reject images unlikely to be fundus photographs.
Preprocessing	Apply CLAHE, resizing, and normalization.
DR Classification	Use EfficientNetB1 transfer learning for five-class severity prediction.
Confidence	Select the highest-probability class and expose its probability as confidence.
Grad-CAM	Generate a heatmap showing regions influencing the predicted class.
Result Display	Present severity, confidence, original image, and explanation.

changes to the trained architecture. This separation also simplifies debugging and testing because individual processing stages can be verified independently.

C. Functional Modules

The implementation is organized into modular components: image upload, image validation, fundus-image verification, image preprocessing, DR classification, prediction and confidence estimation, Grad-CAM explainability, and result display. This separation allows individual modules to be updated or tested independently. The system also specifies validation of uploaded images, rejection of unsupported files, graceful error handling, and modular maintainability.

The modular structure additionally supports future replacement of individual components. For example, the heuristic could later be replaced by a more advanced image-quality or fundus-verification model without requiring a complete redesign of the frontend. Similarly, the classification backbone can be improved independently provided that the expected input and output interfaces are maintained. This makes the proposed architecture suitable as a foundation for subsequent experimentation.

VII. EVALUATION STRATEGY AND DISCUSSION

The project specifies evaluation through accuracy, precision, recall, F1-score, and confusion matrix, together with class-wise analysis for all five DR categories. It also requires unit, integration, image-processing, prediction, explainability, web-application, performance, security, usability, and end-to-end testing. These requirements define an evaluation strategy broader than classification accuracy alone.

For the classifier, accuracy measures the proportion of correctly classified images, while precision and recall provide class-sensitive information. Precision is the proportion of predicted positive instances that are actually positive, while recall is the proportion of actual positive instances that are correctly identified.

The corresponding F1-score combines precision and recall into a single measure that balances the two metrics. A confusion matrix is required to identify systematic confusion between adjacent severity classes, which is particularly relevant in DR grading. Recent surveys also emphasize the importance of evaluating generalization, image quality, and robustness in retinal-image analysis [10], [11], [14].

The input-validation stage should additionally be evaluated using valid fundus images and clearly non-fundus images to determine whether the heuristic behaves as intended. This test is separate from DR classification because its purpose is to identify unsuitable inputs before model inference.

The project does not provide final measured model metrics, training hyperparameters, exact train/test split, or a completed confusion matrix. Consequently, the project reports the evaluation protocol rather than inventing numerical performance. Once experiments are completed, the evaluation section can be populated with the measured class-wise and aggregate results without changing the proposed architecture.

Evaluation should also consider the distinction between classification performance and deployment behavior. A model can produce acceptable classification metrics while still experiencing long inference times, unsuitable memory usage, or failures for unsupported files. Retina-Scan therefore treats response time, input handling, explainability generation, and application stability as additional implementation-level evaluation criteria.

The five-class confusion matrix is particularly useful for identifying whether errors occur primarily between neighboring disease stages. Such analysis can provide more information than overall accuracy because it reveals which severity categories are most frequently confused. Similarly, macro-averaged precision, recall, and F1-score can provide a more balanced view when class frequencies differ. These measures can be reported after completion of the final experiments without changing the system architecture.

A. Testing Requirements

Unit testing is intended to verify image upload, file validation, fundus-image verification, preprocessing, classification, confidence calculation, and Grad-CAM independently. Integration testing checks communication among the frontend, Flask backend, image-validation stage, preprocessing pipeline, EfficientNetB1 model, prediction module, and Grad-CAM module. End-to-end testing follows the complete path from image upload through validation and result display. Performance testing measures image processing, inference, and result-generation response time under normal operating conditions.

Security testing includes file validation, unauthorized-access protection, secure communication in deployment, and protection of uploaded images. The project further requires temporary uploaded images and heatmaps to be securely handled and removed when no longer required. These requirements are important because retinal images and screening outputs can represent sensitive user information.

Testing should include both normal and invalid inputs. Valid retinal images can verify the expected classification path, while unrelated images can verify that the fundus-image verification stage blocks unsuitable inputs. Corrupted files, unsupported extensions, unusually large files, and incomplete requests

should also be handled without terminating the application. Such tests help verify that the system behaves predictably outside the normal inference path.

B. Expected Practical Benefits

The proposed architecture can reduce the time needed for preliminary image assessment by automating the first-stage classification workflow. The fundus-image heuristic adds an initial safeguard by preventing obviously unrelated images from being directly processed by the DR classifier. The browser-based interface also separates the screening interaction from model-development infrastructure. More importantly, the combination of confidence information and Grad-CAM visualization provides two complementary outputs: a quantitative indication of model probability and a spatial explanation of the prediction. The literature supports the general value of transfer learning under data scarcity and the growing need for interpretability in medical AI [3], [5], [10], [17].

From a practical perspective, the system can provide a consistent preliminary workflow for users who need to screen a collection of retinal images before specialist review. The automated result can help organize images according to predicted severity and provide a visual reference for further examination. However, the practical benefit depends on adequate validation, appropriate deployment, and human oversight. The system should therefore be viewed as a tool for assisting the screening process rather than independently determining a patient's clinical condition.

VIII. Limitations and Ethical Considerations

The system identifies several constraints. Prediction quality depends on retinal image quality and on the size, quality, class distribution, and representativeness of the APTOS 2019 dataset. The simple fundus-image heuristic can reject or accept some images incorrectly because it is based on general visual properties rather than a clinically validated fundus-image classifier. The model may also produce incorrect predictions for poor-quality images or cases that differ from the training distribution. EfficientNetB1 inference and Grad-CAM generation require adequate computational resources, and response time can vary with image size, server performance, and hardware.

The system also has a medical limitation: it is a pre-screening and referral-support tool, not a replacement for an ophthalmologist. A confidence score should not be interpreted as a diagnosis, and a Grad-CAM heatmap should not be treated as definitive lesion localization. The application should protect uploaded retinal images, restrict access to authorized users, validate file inputs, and use HTTPS/TLS when deployed remotely. Dataset images must also be used according to their applicable licensing conditions.

Another limitation is dataset shift. Images captured using different cameras or under different acquisition conditions may contain characteristics that are not sufficiently represented in the training data. Differences in patient populations and disease prevalence may also affect observed performance. For this reason, results obtained from the development datasets should not automatically be generalized to all clinical settings.

Ethical deployment also requires clear communication of the system's purpose and limitations. Users should be informed that the output is an automated screening result and not a medical diagnosis. Where a screening workflow is used operationally, appropriate referral procedures and human review should remain available. Privacy controls should limit access to uploaded images and generated outputs, while data retention should follow the requirements of the deployment environment.

Recent retinal-image research similarly identifies image quality, domain variation, interpretability, and generalization as important challenges for clinical deployment [11], [14], [18]. These considerations reinforce the need for external validation before real-world use.

IX. Future Scope

The project proposes four principal extensions. First, cross-population validation can be performed using diverse datasets to study generalization across patient demographics and imaging equipment. Second, continuous learning can be introduced by periodically retraining the model with newly labeled retinal images. Third, integration with hospital Electronic Health Record (EHR) systems could connect screening results with patient records and referral workflows. Fourth, the pipeline can be extended to other retinal diseases such as glaucoma and age-related macular degeneration using appropriate datasets and models.

Additional research can investigate calibration of confidence scores, domain adaptation across cameras and institutions, more rigorous external validation, improved fundus-image quality and validity detection, and clinician-centered evaluation of Grad-CAM explanations. These directions are consistent with literature observations that dataset shift, interpretability, transferability, and broad retinal screening remain important research challenges [2], [5], [8], [11], [14]. Recent multi-level and broad-coverage fundus classification studies also suggest opportunities for extending the system beyond a single retinal condition [15], [18].

Future development can also improve the input-validation stage by replacing the rule-based heuristic with a dedicated fundus-quality or fundus-verification model trained on suitable examples. Such an extension could provide a more systematic method for identifying blurred, severely underexposed, overexposed, or non-retinal images. Similarly, confidence calibration could be investigated so that the displayed probability better reflects the observed likelihood of correct predictions.

Another possible extension is deployment optimization. Model compression, efficient inference, and hardware-aware optimization could reduce the computational cost of web-based screening. A more comprehensive logging and monitoring mechanism could also record system-level failures and processing times without retaining unnecessary patient data. These improvements would support a more robust transition

from a project prototype to a controlled screening-support environment.

X. Conclusion

Retina-Scan presents a web-based deep learning system for pre-screening and referral support in Diabetic Retinopathy. The design combines CLAHE-based image enhancement, EfficientNetB1 transfer learning implemented using PyTorch, class-weighted training, confidence estimation, and Grad-CAM visualization within a Flask-backed web application. An additional simple fundus-image heuristic is used to check whether an uploaded image is likely to be a retinal fundus image before performing DR classification. The five-class formulation covers No DR, Mild, Moderate, Severe, and Proliferative DR.

The proposed system is designed to make preliminary screening more accessible while retaining a clear clinical limitation: its output is decision support and not a definitive medical diagnosis. The current project defines the required evaluation metrics and testing procedures but does not provide final experimental measurements; therefore, the project reports the evaluation framework rather than unsupported accuracy claims. Future validation using the project datasets and diverse external data will be necessary before any broader clinical conclusions can be drawn.

The complete pipeline also demonstrates how deep-learning inference can be integrated with practical software components to form a usable screening application. Rather than presenting only a classification model, Retina-Scan combines input verification, preprocessing, disease grading, confidence reporting, explainability, and web-based presentation. This provides a structured basis for future experimental evaluation and system refinement.

In its current form, the project establishes the architectural and methodological foundation for an automated retinal screening prototype. The next stage should focus on completing controlled experiments, reporting class-wise performance, examining failure cases, and validating the system on data that differ from the training distribution. Such validation is essential for determining the reliability and practical usefulness of the proposed approach.

Furthermore, the modular architecture allows individual components to be improved without redesigning the entire application. Future improvements in model performance, image-quality verification, and result presentation could strengthen its usefulness as a screening-support tool. Ultimately, clinical evaluation and professional oversight will remain essential before the system can be considered for practical healthcare deployment.

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